DoangJoo "Alan" Synn

PhD Candidate in Computer Science · Georgia Institute of Technology
Advised by Prof. HyunJoo Oh and Prof. Sehoon Ha
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Research Interests

I study **motion** as a computational medium across HCI and computer graphics. My research builds systems for designing, generating, and fabricating motion. I aim to make motion programmable and tangible across animated characters, mechanical artifacts, robotic systems, and AI-assisted creative tools.

Education

Georgia Tech, Ph.D. in Computer Science *2022 – Present*

- Atlanta, GA, USA
- School of Interactive Computing; advised by Profs. Oh & Ha
- Research: computational design, automata, and motion synthesis (HCI/Graphics)

Korea University, M.Eng. in ECE *2020 – 2022*

- Seoul, Korea
- Advised by Prof. Jong-Kook Kim
- GPA: 4.25 (4.5 scale)
- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

Korea University, B.Eng. in EE *2012 – 2020*

- Seoul, Korea
- GPA: 3.56 (4.5 scale)
- Military service leave as a Software Engineer, 2016–2019

Experience

PhD Research Intern, Dolby Laboratories *2024 – 2025*

- Atlanta, GA, USA
- Experience Delivery Lab, built a time-series, multimodal datastore supporting low-latency media analytics.

MLOps Engineering Intern, LG Uplus Corp. *2023*

- Seoul, Korea
- Led **DeepCrunch**, a model-compression library with multi-backend AI inference for efficient edge serving.

Research Scientist, Protopia AI Inc. *2021 – 2022*

- Austin, TX, USA
- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.
- CEO: Dr. Ebrahimi; CTO: Prof. Esmaeilzadeh.

Cloud Architect · AWS Educate Cloud Ambassador *2020 – 2022*

- Seoul, Korea
- Led engineering for 4 startup services, translating ambiguous product needs into back-end systems while mentoring early-career engineers.
- Built production cloud infrastructure with Kubernetes, Terraform, CI/CD, and cost-optimized autoscaling.

Server Engineer, Flysher Inc. *2019*

- Seoul, Korea
- Shipped back-end features for 4 social-game titles; one reached global top-10 on Facebook.
- Led an ETL ⟶ Redshift analytics pipeline driving data-informed feature rollout.

System Engineer, AKA Intelligence (Musio) *2017 – 2019*

- Seoul, Korea
- Designed embedded Linux boards & firmware for the social robot **Musio**.
- Coordinated HW-SW integration and production QA with overseas ODMs.

Selected Publications

1. **D. Synn**, Z. Guo, S. Ha, and H. Oh, "MotionSmith: A Sketch-Based Design System for Automata Making," *CHI*, Apr. 2026.
2. J. Yu, P. Garg, **D. Synn**, and H. Oh, "Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning," *CHI*, Apr. 2025.
3. M. Ryu, X. Piao, J. Park, **D. Synn**, and J. Kim, "ADaPT: An Automated Dataloader Parameter Tuning Framework using AVL Tree-based Search Algorithms," *TKIPS*, Jan. 2025.
4. **D. Synn**, X. Piao, J. Park, and J. Kim, "Distributed Training of Deep Learning Models Using Micro Batch Streaming," *KTSDE*, Mar. 2023.
5. X. Piao, **D. Synn**, J. Park, and J. Kim, "Enabling large batch size training for dnn models beyond the memory limit while maintaining performance," *IEEE Access*, 2023.
6. J. Han, **D. Synn**, T. Kim, H. Jung, and J. Kim, "Feature Based Sampling: A Fast and Robust Sampling Method for Tasks Using 3D Point Cloud," *IEEE Access*, Apr. 2022.
7. **D. Synn** and J. Kim, "Num Worker Tuner: An Automated Spawn Parameter Tuner for Multi-Processing DataLoaders," *ACK*, Oct. 2021.
8. J. Park, **D. Synn**, and J. Kim, "A Survey on the Advancement of Virtualization Technology," *KIPS*, May 2021.
9. B. Ahn, **D. Synn**, M. Derkani, E. Ebrahimi, and H. Esmaeilzadeh, "Protopia AI: Taking on the Missing Link in AI Privacy and Data Protection," *NeurIPS*, 2021.
10. D. Kim, **D. Synn**, and J. Kim, "SG-Drop: Faster Skip-Gram by Dropping Context Words," *KIPS*, Nov. 2020.